# ANDREA ADELFIO

**Data Scientist | ML Engineer | Software Engineer**


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## TECH SKILLS

### Programming

Python (Expert) | SQL (Expert) | Java (Proficient) | Bash | JavaScript | C/C++ | LaTeX

### Machine Learning & AI

Deep Learning | Bayesian Neural Networks | Anomaly Detection | Time Series Analysis | TensorFlow, PyTorch, Keras

### Data Engineering & DevOps

Docker | Kubernetes | Git | CI/CD | HPC | Data Lakes | Atlassian (Jira, Confluence...)

## EDUCATION

M.Sc. in Physics (EQF 7)
**University of Trieste**
2018 – 2020

Thesis: *Gravitational Wave Counterparts Detection with CTA.*
Final grade: 101/110.

B.Sc. in Physics (EQF 6)
**University of Cagliari**
2015 – 2018

Final grade: 97/110.

## LANGUAGES

-**Italian** (Mother tongue)
-**English** (C2 – Cambridge C2 Proficiency, score 202)
-**German** (B1)

## CERTIFICATIONS & COURSES

-**School on Open Science Cloud** (kw: MLOps, Docker, Kubernetes)
-**Container Fundamentals for Scientific Research** (kw: Docker, Kubernetes)
-**High Performance Computing** (kw: HPC, c++, openmp)
-**Uncertainty Quantification for Machine Learning**
-**FPGA Programming**

## PROFESSIONAL PROFILE

*Profile with five years of dual experience as Data Scientist, with a background in Artificial Intelligence and Machine Learning applied to data and time series analysis, and as Software Developer with expertise in Agile methodologies (Scrum) and CI/CD practices.*
*Ability to work across research and industrial contexts, with direct involvement in machine learning model development, software engineering and data infrastructures, in domains from the banking sector to satellite monitoring.*

## WORK EXPERIENCE

Data Scientist & ML Engineer
**Italian Research Center on High Performance Computing, Big Data and Quantum Computing (ICSC) & National Institute of Nuclear Physics (INFN)**
Sep 2023 – Dec 2025 | Perugia, Italy

- **ML Software Development:** Design, development and maintenance of TSLies (link), an open-source Python framework for neural-network-based anomaly detection (FFNN, RNN, Bayesian Neural Networks) on multivariate time-series data (TensorFlow and TensorFlow-Probability -based pipeline). Applied to Fermi (NASA) satellite data.
- **Mentoring & Technical Leadership:** Supervision and mentoring of junior researchers and Master's students, with guidance on software engineering best practices and code optimization. Contribution to two Master Classes for programming laboratories.
- **Industrial Impact:** Implementation of the ATS framework for financial anomaly monitoring for Intesa Sanpaolo, bridging research and industry requirements.
- **Data Infrastructure:** Design of scalable Data Lakes and storage architectures for the Italian Computing Center infrastructure, ensuring reliability and reproducibility for large-scale datasets.

Software Engineer
**Accenture Technology Solutions**
May 2021 – Aug 2023 | Milan, Italy

- **Full-Stack Engineering:** Development and maintenance of mission-critical enterprise software using Java, SQL and JavaScript.
- **Production Deployment:** Management of end-to-end software lifecycles, including version control, unit testing and deployment to production environments.
- **Agile & DevOps:** Work within Scrum teams, contributing to CI/CD pipelines and workflow automation.
- **Consultancy:** Delivery of scalable technical solutions to international clients.

## KEY PROJECTS

**TSLies – Time Series anomaLies**
Author and maintainer of an open-source Python package for anomaly detection in time series data based on neural networks and statistical learning methods.

**Satellite Anomaly Detection**
Design and implementation of modular ML pipelines for transient and anomaly detection on large-scale satellite time series data.

**Data Infrastructure**
Development of data-access, preprocessing pipelines and storage solutions for large datasets with reproducible workflows.

## PUBLICATIONS

- "**Advances on TSLies: Time Series AnomaLIES, an Efficient Ma- chine Learning Based Anomaly Detection Framework**". Andrea Adelfio et al. in: Astronomy and Computing (2025, under review).
- "**Scalable and Interoperable Data Management in the Spoke 3 Big Data Infrastructure**". Giacomo Coran et al. in: Astronomy and Computing (2025, under review)
- "**Anomaly Detection with Machine Learning on Time Series: Unveiling Lost Transients Data**". Andrea Adelfio et al., in: Proceedings of the Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP). 2025.
- "**Spoke 3 Big Data Management, Storage, and Archive Infrastructure**". Giacomo Coran et al. in: Proceedings of the Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP). 2025.
- "**Searching for Very-High-Energy Electromagnetic Counterparts to Gravitational-Wave Events with the Cherenkov Telescope Array**". Barbara Patricelli et al. in: Proceedings of the 37th International Cosmic Ray Conference. 2021.